C Programming Core Practice Problems

A structured compilation of fundamental logic-building exercises focusing on loops, conditionals, and arithmetic manipulation.

Core Logic & Arithmetic Problems

1. Sum of digits	Write a program to find the sum of all digits of a given number.
2. Reverse a number	Reverse the digits of a given number (e.g., 1234 becomes 4321).
3. Count digits	Count how many digits are present in a number, handling edge cases like 0 properly.
4. Palindrome number	Check whether a given number reads the same forwards and backwards (e.g., 121).
5. Even or odd	Check whether a given integer is even or odd.
6. Prime number	Determine whether a number is prime (only divisible by 1 and itself).
7. Factorial	Find the factorial of a number using an iterative loop structure (n!).
8. Fibonacci series	Print the first n terms of the Fibonacci sequence starting from 0 and 1.
9. Sum of first n numbers	Calculate the sum of the first n natural numbers (1 + 2 + 3 + ... + n).
10. Multiplication table	Print the complete multiplication table of a given integer.
11. Largest of three numbers	Find the maximum among three numbers using conditional statements.
12. Smallest of three numbers	Find the minimum among three numbers using conditional statements.
13. Power of a number	Calculate the value of a raised to the power of b (a^b) manually without using the math.h pow() function.
14. Armstrong number	Check whether a number is equal to the sum of cubes of its digits (for 3-digit numbers).
15. Perfect number	Check if a number is equal to the sum of its proper positive divisors, excluding itself.
16. GCD / HCF	Find the Greatest Common Divisor (Highest Common Factor) of two numbers using Euclidean algorithm or loops.

17. LCM	Find the Least Common Multiple of two numbers based on their GCD relationship.
18. Prime numbers in a range	Print all prime numbers falling between a user-defined lower and upper bound interval.
19. Sum of even digits	Isolate and find the sum of only the even-valued digits within a number.
20. Sum of odd digits	Isolate and find the sum of only the odd-valued digits within a number.

Advanced Logic & Skill Builders

21. Largest digit in a number	Scan through all digits of an integer to find the maximum face value.
22. Smallest digit in a number	Scan through all digits of an integer to find the minimum face value.
23. Count even and odd digits	Simultaneously count how many digits in a number are even and how many are odd.
24. Strong number check	Check if a number equals the sum of the factorials of its individual digits (e.g., 145 = 1! + 4! + 5!).
25. Fibonacci up to a limit	Print all terms of the Fibonacci sequence that fall below or equal a specified maximum limit value.
26. Find the nth Fibonacci number	Directly compute and output only the nth term of the Fibonacci sequence.
27. Decimal to binary conversion	Convert a base-10 integer into its corresponding base-2 binary representation.
28. Binary to decimal conversion	Convert a base-2 binary stream or integer into its corresponding base-10 value.
29. Sum of first and last digit	Isolate the most significant digit and least significant digit of a number and find their sum.
30. Truncate last digit	Remove the trailing digit of an integer and display the remaining sequence of numbers.